



Tribhuvan University

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A Lab Report

On

Research Methodology

Submitted by

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Q1: Design a Research Questionnaire for data collection

Research Topic “Use of AI Tools among Computer Application Students”

Problem Statement

To study the usage, awareness, and satisfaction of AI tools among Computer Application students.

Research Objectives

1. To identify the commonly used AI tools among Computer Application students.
2. To examine the frequency of AI tool usage.
3. To measure students' satisfaction with AI tools.
4. To identify the perceived benefits and challenges of using AI tools.

Questionnaire

Title

Questionnaire on the Use of AI Tools among Computer Application Students

Introduction

Dear Respondent,

This questionnaire is designed to collect information about the use of Artificial Intelligence (AI) tools among Computer Application students. The information collected will be used for academic research purposes only. Your responses will be kept confidential.

Please select the most appropriate answer.

Section A: Demographic Information

1. Gender:

- ☐ Male
- ☐ Female
- ☐ Other
- ☐ Prefer not to say

2. Age Group:

- ☐ Below 18
- ☐ 18–20
- ☐ 21–23
- ☐ Above 23

3. Academic Level:

- ☐ Bachelor
- ☐ Master
- ☐ Other

4. How long have you been using AI tools?

- ☐ Less than 6 months
- ☐ 6 months–1 year
- ☐ 1–2 years
- ☐ More than 2 years

Section B: AI Tool Usage

5. Which AI tool do you use most frequently?

- ☐ ChatGPT
- ☐ Google Gemini
- ☐ Microsoft Copilot
- ☐ Claude
- ☐ Other

6. How frequently do you use AI tools?

- ☐ Daily
- ☐ Several times a week
- ☐ Once a week
- ☐ Rarely
- ☐ Never

7. What do you mainly use AI tools for?

- ☐ Programming/Coding
- ☐ Assignment/Research
- ☐ Content Writing
- ☐ Learning/Study
- ☐ Other

8. Have AI tools helped you complete academic tasks?

- ☐ Yes
- ☐ No

Section C: Satisfaction and Perception

9. AI tools are easy to use.

Response	Select
Strongly Disagree	☐
Disagree	☐
Neutral	☐
Agree	☐
Strongly Agree	☐

10. AI tools improve my learning efficiency.

Response	Select
Strongly Disagree	☐
Disagree	☐

Response	Select
Neutral	☐
Agree	☐
Strongly Agree	☐

11. AI tools help me understand programming concepts better.

Response	Select
Strongly Disagree	☐
Disagree	☐
Neutral	☐
Agree	☐
Strongly Agree	☐

12. Overall, how satisfied are you with AI tools for academic purposes?

Rating:

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

Where:

1 = Very Dissatisfied, 2 = Dissatisfied, 3 = Neutral, 4 = Satisfied, 5 = Very Satisfied

Section D: Open-Ended Question

13. What are the major benefits or challenges you experience while using AI tools for academic purposes?

Variables Used

- **Independent variables:** Age, academic level, duration of AI usage, frequency of usage, type of AI tool.
- **Dependent variables:** Satisfaction, perceived learning efficiency, perceived usefulness.
- **Demographic variables:** Gender, age, academic level.

Conclusion

The questionnaire collects primary data about **AI tool usage, frequency, purpose, satisfaction, perceived usefulness, and challenges** among Computer Application students. It includes demographic, multiple-choice, Likert-scale, Yes/No, rating-scale, and open-ended questions as required by the practical.

Q2: Questionnaire Validation

Research Topic: Use of AI Tools among Computer Application Students

Questionnaire validation is performed using the questionnaire developed in Question 1. The purpose is to check whether the questionnaire is clear, relevant, non-repetitive, and suitable for collecting the required research data. The practical requires identifying variables, classifying them, explaining sections, conducting a pilot survey, identifying problems, and revising the questionnaire.

1. Variables Being Measured

Variable	Measurement
Gender	Demographic
Age Group	Demographic
Academic Level	Demographic
Duration of AI usage	Usage
Type of AI tool	Usage
Frequency of AI usage	Usage
Purpose of AI usage	Usage
Academic task assistance	Usage
Ease of use	Perception
Learning efficiency	Perception
Programming understanding	Perception
Overall satisfaction	Satisfaction
Benefits and challenges	Open-ended

2. Classification of Variables

Variable	Type of Scale
Gender	Nominal
Age Group	Ordinal
Academic Level	Nominal
Duration of AI usage	Ordinal
Type of AI tool	Nominal
Frequency of usage	Ordinal
Purpose of usage	Nominal
Academic task assistance	Nominal
Ease of use	Ordinal
Learning efficiency	Ordinal
Programming understanding	Ordinal
Satisfaction rating (1–5)	Ordinal
Benefits/Challenges	Qualitative

Note: Likert-scale responses are treated as **ordinal** variables in this practical.

Q3. Purpose of Each Section

Section A: Demographic Information

Collects basic information about respondents such as gender, age, and academic level.

Section B: AI Tool Usage

Measures the type, frequency, duration, and purpose of AI tool usage.

Section C: Satisfaction and Perception

Measures students' perceptions, learning benefits, ease of use, and satisfaction using Likert and rating scales.

Section D: Open-Ended Question

Allows respondents to provide their own opinions about the benefits and challenges of AI tools.

4. Pilot Survey

A small pilot survey is conducted before the final data collection.

Sample: 10 Computer Application students

Procedure:

1. Select 10 students for the pilot survey.
2. Provide them with the questionnaire.
3. Ask them to complete all questions.
4. Ask whether any question is confusing, repetitive, difficult, or biased.
5. Record their feedback.
6. Revise the questionnaire accordingly.

5. Problems Identified from Pilot Survey

Problem	Example	Action Taken
Ambiguous question	"How often do you use AI?"	Added specific frequency options
Overlapping options	Similar academic purposes	Combined/clarified options
Unclear terminology	"AI tools"	Provided examples such as ChatGPT, Gemini
Rating ambiguity	Rating 1–5 without meaning	Defined 1 and 5
Open-ended question	Too broad	Clearly asked about benefits/challenges

6. Revised Questionnaire

After the pilot survey, the questionnaire was revised by:

- Clarifying ambiguous questions.
- Removing unnecessary/repetitive questions.
- Providing clear response options.
- Defining the meaning of the 1–5 rating scale.
- Making the questions neutral and unbiased.
- Ensuring that questions are related to the research objectives.

7. Conclusion

The questionnaire was validated through **variable identification, scale classification, section review, pilot testing, identification of ambiguous/repetitive/biased questions, and revision**. The revised questionnaire is therefore suitable for collecting primary data on the use of AI tools among Computer Application students.

Section B: Research Proposal Using LaTeX Question 3: Prepare a Research Proposal Using LaTeX

Educational Applications.

Below is a practical-exam-ready LaTeX document using that topic.

```
\documentclass[12pt,a4paper]{article}

% Packages
\usepackage[utf8]{inputenc}
\usepackage{geometry}
\usepackage{setspace}
\usepackage{graphicx}
\usepackage{booktabs}
\usepackage{hyperref}

\geometry{margin=1in}
\onehalfspacing

\title{\textbf{Research Proposal}\\
\vspace{0.5cm}
\textbf{User Acceptance of AI-Based Educational Applications Among University Students}}

\author{
Bishal Rijal\\
MCA, Tribhuvan University
}

\date{\today}

\begin{document}

\maketitle

\tableofcontents
\newpage

%-----
\section{Introduction}
```

Artificial Intelligence (AI) has become an important technology in modern education. AI-based educational applications provide students with personalized learning, automated assistance, content generation, and interactive learning opportunities.

University students increasingly use AI-based applications for learning, programming, research, assignment preparation, and solving academic problems. However, the acceptance and effective use of these applications may depend on factors such as perceived usefulness, ease of use, trust, and

students' previous experience with technology.

This study aims to investigate the factors affecting university students' acceptance of AI-based educational applications.

%-----

\section{Background of the Study}

AI-based educational applications include intelligent tutoring systems, AI chatbots, automated assessment tools, recommendation systems, and generative AI applications.

These applications can support students by providing quick feedback, personalized assistance, and access to learning resources. Despite these potential benefits, students may have concerns regarding accuracy, privacy, reliability, academic integrity, and dependence on AI.

Therefore, understanding students' acceptance of AI-based educational applications is important for effective implementation in higher education.

%-----

\section{Problem Statement}

Although AI-based educational applications are increasingly available to students, their acceptance and usage may vary among university students.

Some students may find AI applications useful and easy to use, while others may have concerns about accuracy, privacy, reliability, and academic integrity.

Therefore, this research focuses on identifying the factors that influence students' acceptance of AI-based educational applications.

%-----

\section{Research Objectives}

The main objective of this research is to study the acceptance of AI-based educational applications among university students.

The specific objectives are:

\begin{enumerate}

\item To identify the level of acceptance of AI-based educational applications among university students.

\item To examine the effect of perceived usefulness on students'

acceptance of AI applications.

\item To examine the effect of perceived ease of use on students' acceptance.

\item To identify students' major concerns regarding AI-based educational applications.

\end{enumerate}

%-----

\section{Research Questions}

\begin{enumerate}

\item What is the level of acceptance of AI-based educational applications among university students?

\item Does perceived usefulness influence students' acceptance of AI-based educational applications?

\item Does perceived ease of use influence students' acceptance?

\item What are the major concerns of students regarding AI-based educational applications?

\end{enumerate}

%-----

\section{Research Hypothesis}

The following hypotheses will be tested:

\begin{itemize}

\item \$H_0\$: Perceived usefulness has no significant relationship with students' acceptance of AI-based educational applications.

\item \$H_1\$: Perceived usefulness has a significant relationship with students' acceptance of AI-based educational applications.

\item \$H_0\$: Perceived ease of use has no significant relationship with students' acceptance of AI-based educational applications.

\item \$H_1\$: Perceived ease of use has a significant relationship with students' acceptance of AI-based educational applications.

\end{itemize}

%-----

\section{Significance of the Study}

The study may be useful to:

\begin{itemize}

\item University students for understanding the effective use of

AI-based educational applications.

\item Teachers and educators for understanding students' acceptance of AI technologies.

\item Educational institutions for planning AI-supported learning activities.

\item Researchers for conducting further research on AI in education.

\end{itemize}

%-----

\section{Scope and Limitations}

The study focuses on university students and their acceptance of AI-based educational applications.

The study is limited to selected university students and may not represent all students. The findings will also depend on the accuracy and honesty of the responses provided by participants.

%-----

\section{Literature Review}

Previous research on educational technology indicates that perceived usefulness and perceived ease of use are important factors in technology acceptance.

AI-based educational applications provide students with opportunities for personalized learning, automated assistance, and interactive learning. However, concerns related to privacy, reliability, accuracy, and academic integrity may affect students' willingness to use such applications.

This study will review relevant research articles to identify existing findings and research gaps related to students' acceptance of AI-based educational applications.

%-----

\section{Research Methodology}

This study will use a quantitative research approach. Primary data will be collected from university students using a structured questionnaire.

The questionnaire will contain demographic questions, multiple-choice questions, Yes/No questions, Likert-scale questions, rating-scale questions, and open-ended questions.

%-----

\section{Research Design}

A descriptive and analytical research design will be used.

The descriptive component will describe students' AI usage and acceptance, while the analytical component will examine relationships between variables such as perceived usefulness, perceived ease of use, and acceptance.

%-----
\section{Population and Sample}

The population of the study will consist of university students who use or have experience with AI-based educational applications.

A sample of selected university students will be taken for the study.

%-----
\section{Sampling Technique}

Convenience sampling will be used to select respondents who are available and willing to participate in the study.

%-----
\section{Data Collection Methods}

Primary data will be collected using a structured questionnaire.

The questionnaire will be distributed to selected university students through online or physical forms. Respondents will be informed that the collected information will be used only for academic research purposes.

%-----
\section{Data Analysis Techniques}

The collected data will be analyzed using statistical software such as Python, R, SPSS, or other relevant tools.

The following techniques may be used:

\begin{itemize}
 \item Frequency and percentage
 \item Mean and standard deviation
 \item Correlation analysis
 \item Chi-square test
 \item Appropriate hypothesis testing
\end{itemize}

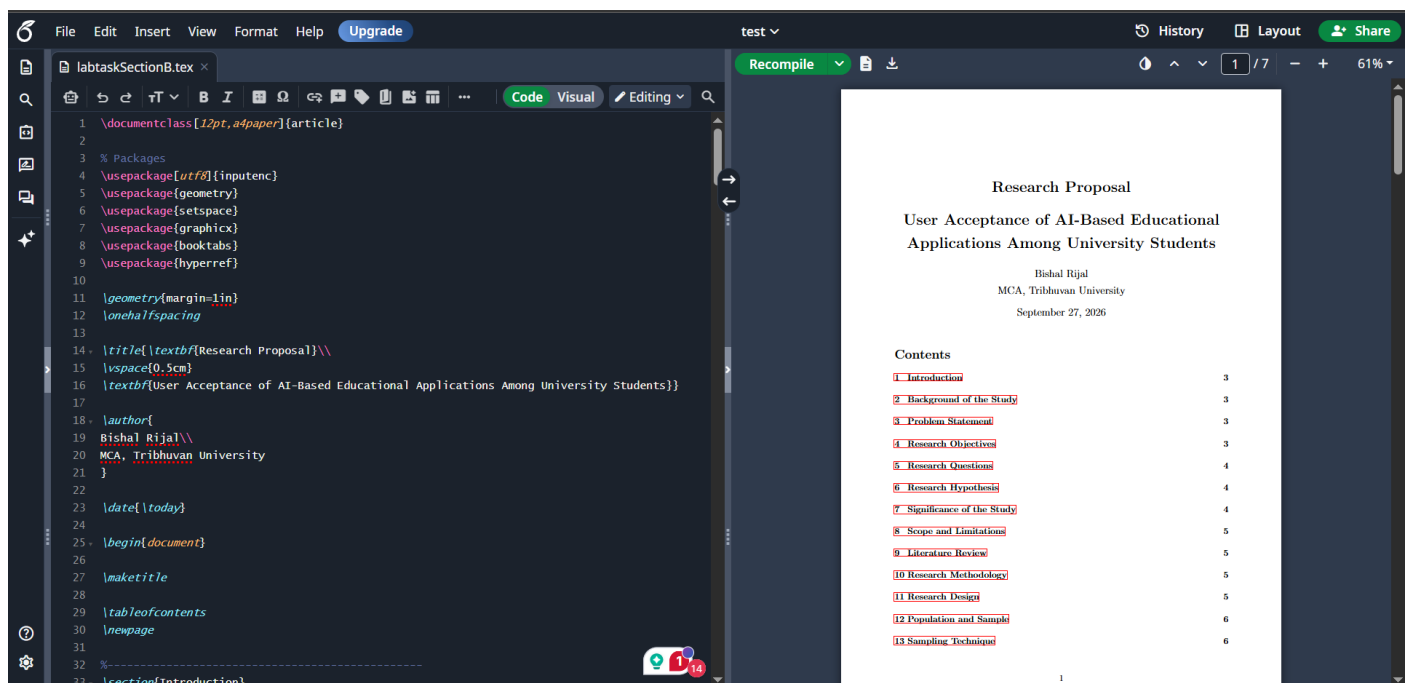
(2003). User acceptance of information technology: Toward a unified view. MIS Quarterly.

Relevant research articles on AI-based educational applications, technology acceptance, and AI in higher education will be reviewed during the final research study.

Practical Procedure

1. Open **Overleaf** or another LaTeX editor.
2. Create a new LaTeX project.
3. Create/open main.tex.
4. Copy and paste the above code.
5. Compile the document.
6. Check the generated PDF.
7. Submit/print the research proposal as required.

The proposal contains all the major sections specifically listed in the practical question, including **introduction, background, problem statement, objectives, research questions, hypothesis, significance, scope, literature review, methodology, research design, population/sample, sampling, data collection, analysis, expected outcomes, timeline, and references.**



SECTION C: LITERATURE REVIEW AND CITATION

Q4. Literature Review Using Mendeley/Zotero

Research Topic

User Acceptance of AI-Based Educational Applications Among University Students

Procedure

1. Open **Mendeley/Zotero**.
2. Create a new folder named:
AI Educational Applications.
3. Search for at least 10 relevant research articles from academic sources.
4. Download the research papers.
5. Import the papers into Mendeley/Zotero.
6. Organize the papers using folders/tags.
7. Read each paper.
8. Record:
 - Research objective
 - Methodology
 - Dataset/sample
 - Findings
 - Limitations
9. Identify the research gap.
10. Insert in-text citations.
11. Generate the reference list.
12. Select **APA or IEEE** citation style.
13. Prepare the literature review.

Literature Review Table

Paper	Objective	Methodology	Finding	Limitation
Paper 1	Study AI acceptance	Survey	AI usefulness affects acceptance	Limited sample
Paper 2	Study AI in education	Quantitative	AI supports learning	Limited institution
Paper 3	Study technology adoption	Survey	Ease of use is important	Self-reported data
Paper 4	Study AI usage	Survey	Students use AI for learning	Small sample
Paper 5	Study student perception	Questionnaire	Positive perception	Limited location
Paper 6	Study AI chatbot usage	Quantitative	Chatbots support learning	Limited users
Paper 7	Study AI acceptance	Survey	Usefulness affects intention	Cross-sectional
Paper 8	Study AI learning	Experimental	AI provides learning support	Short duration
Paper 9	Study AI concerns	Survey	Privacy is a concern	Limited sample
Paper 10	Study educational AI	Review	AI has educational applications	More research needed

Research Gap

Existing studies have examined AI adoption and educational applications, but further research can examine the acceptance of AI-based educational applications among university students in a specific local context.

Output

- Mendeley/Zotero library
- Organized research papers
- Literature review
- In-text citations
- Automatically generated bibliography

The practical specifically requires at least **10 research articles** and identification of objectives, methodologies, datasets, findings, limitations, and research gaps.

Q5. Citation and Referencing

Objective

To prepare a research document containing at least 10 in-text citations and 10 references using Mendeley/Zotero.

Procedure

1. Open Mendeley/Zotero.
2. Import at least 10 academic sources.
3. Create a research document in MS Word/LaTeX.
4. Select citation style such as APA or IEEE.
5. Insert at least 10 in-text citations.
6. Insert citations at appropriate locations.
7. Generate the bibliography automatically.
8. Verify the references.
9. Check citation consistency.
10. Save the final document.

Example APA In-Text Citations

Technology acceptance is influenced by perceived usefulness and perceived ease of use (Davis, 1989).

The Unified Theory of Acceptance and Use of Technology provides a framework for understanding technology acceptance (Venkatesh et al., 2003).

Example References

Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*.

Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *MIS Quarterly*.

Output:

- Minimum 10 in-text citations
- Minimum 10 references

- Different source types
- Automatically generated bibliography
- Consistent citation style

These are the requirements specified in the practical sheet.

SECTION D: PLAGIARISM CHECKING

Q6. Plagiarism Checking Using Turnitin

Objective

To check the similarity of a research proposal/report using Turnitin.

Procedure

Step 1: Prepare Research Document

Prepare a research proposal/report of at least **1500 words** on a Computer Application-related topic.

Step 2: Upload Document

1. Login to Turnitin.
2. Open the assigned submission.
3. Upload the research document.
4. Submit the document.

Step 3: Generate Similarity Report

Open the generated similarity report and examine highlighted sections.

Step 4: Identify Problems

Check for:

- Improper citations
- Direct copying
- Insufficient paraphrasing
- Missing references

Step 5: Revise

Rewrite improperly paraphrased content and add appropriate citations.

Step 6: Resubmit

Upload the revised document if resubmission is permitted.

Step 7: Compare Reports

Compare the original and revised similarity reports.

Output

Submit:

1. Research proposal/report
2. Original similarity report
3. Revised similarity report
4. Brief explanation of changes made

The practical explicitly requires these steps and outputs.

SECTION E: STATISTICAL DATA ANALYSIS

Q7. Descriptive Statistics

Given Data

10, 30, 50, 40, 20, 60, 70, 80, 90, 31, 25, 30

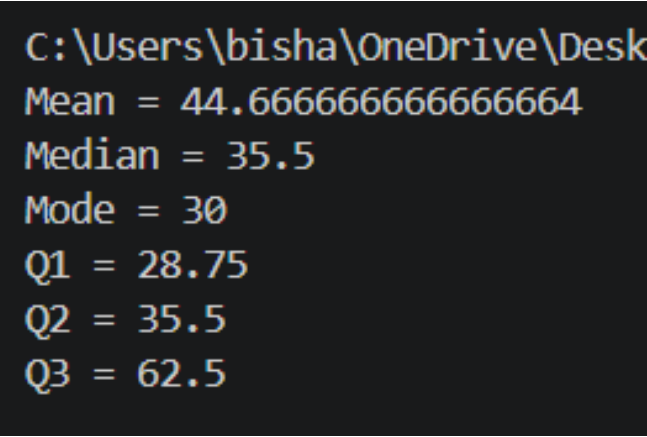
```
import numpy as np
from scipy import stats

data = [10, 30, 50, 40, 20, 60, 70, 80, 90, 31, 25, 30]

mean = np.mean(data)
median = np.median(data)
mode = stats.mode(data, keepdims=True).mode[0]

q1 = np.percentile(data, 25)
q2 = np.percentile(data, 50)
q3 = np.percentile(data, 75)

print("Mean =", mean)
print("Median =", median)
print("Mode =", mode)
print("Q1 =", q1)
print("Q2 =", q2)
print("Q3 =", q3)
```



```
C:\Users\bisha\OneDrive\Desktop
Mean = 44.666666666666664
Median = 35.5
Mode = 30
Q1 = 28.75
Q2 = 35.5
Q3 = 62.5
```

Results

Measure	Value
Mean	44.67
Median	35.50
Mode	30
Q1/P25	28.75
Q2/P50	35.50
Q3/P75	62.50

Interpretation

The average examination mark is approximately **44.67**. The median is **35.5**, and the most frequently occurring mark is **30**.

The middle 50% of observations lie approximately between **28.75 and 62.50**.

Q8. Chi-Square Test of Independence

Given Data

Gender	Prefer	Do Not Prefer	Total
Male	45	25	70
Female	35	15	50
Total	80	40	120

Hypotheses

H₀: Gender and preference for an online learning application are independent.

H₁: Gender and preference for an online learning application are associated.

Expected Frequency Formula

$$E = \frac{\text{Row Total} \times \text{Column Total}}{\text{Grand Total}}$$

Expected Frequencies

Male–Prefer:

$$E = \frac{70 \times 80}{120} = 46.67$$

Male–Do Not Prefer:

$$E = \frac{70 \times 40}{120} = 23.33$$

Female–Prefer:

$$E = \frac{50 \times 80}{120} = 33.33$$

Female–Do Not Prefer:

Gender	Prefer	Do Not Prefer
Male	46.67	23.33
Female	33.33	16.67

Python Code

```
import numpy as np
from scipy.stats import chi2_contingency
```

```
data = np.array([
    [45, 25],
    [35, 15]
])
```

```
chi2, p, df, expected = chi2_contingency(data, correction=False)
```

```
print("Chi-Square =", chi2)
print("Degrees of Freedom =", df)
print("p-value =", p)
print("Expected Frequencies:")
print(expected)
```

```
C:\Users\bisha\OneDrive\Desktop\MC
Chi-Square = 0.4285714285714285
Degrees of Freedom = 1
p-value = 0.5126907602619235
Expected Frequencies:
[[46.66666667 23.33333333]
 [33.33333333 16.66666667]]
```

At 5% significance level:

$$\alpha = 0.05$$

Since:

$$p = 0.5127 > 0.05$$

we **fail to reject H_0** .

Conclusion

There is **no statistically significant association** between gender and preference for the online learning application at the 5% significance level.

Q9. Spearman Rank Correlation

Given Data

Student	X	Y
1	25	58
2	19	52
3	28	65
4	30	70
5	18	51
6	24	62

Hypotheses

H₀: There is no significant relationship between application usage hours and student performance.

H₁: There is a significant relationship between application usage hours and student performance.

Python Code

```
from scipy.stats import spearmanr
```

```
X = [25, 19, 28, 30, 18, 24]
```

```
Y = [58, 52, 65, 70, 51, 62]
```

```
rho, p = spearmanr(X, Y)
```

```
print("Spearman Correlation =", rho)
```

```
print("p-value =", p)
```

```
C:\Users\bisha\OneDrive\Desktop\MCA\First s  
Spearman Correlation = 0.942857142857143  
p-value = 0.004804664723032055
```

Interpretation

$$\rho = 0.9429$$

This represents a **strong positive relationship** between the number of hours spent using the educational application and student performance.

Since:

$$p = 0.0048 < 0.05$$

the null hypothesis is rejected.

Conclusion

There is a statistically significant positive relationship between educational application usage hours and student performance in the given sample.

Q10. Independent Samples T-Test

Data

Method A

65, 70, 68, 75, 72, 69, 71, 73

Method B

72, 75, 70, 78, 80, 74, 77, 76

Hypotheses

H₀: There is no significant difference between the mean scores of Method A and Method B.

H₁: There is a significant difference between the mean scores.

Python Code

```
import numpy as np
from scipy.stats import ttest_ind

A = np.array([65,70,68,75,72,69,71,73])
B = np.array([72,75,70,78,80,74,77,76])

mean_A = np.mean(A)
mean_B = np.mean(B)

sd_A = np.std(A, ddof=1)
sd_B = np.std(B, ddof=1)

t, p = ttest_ind(A, B, equal_var=True)

print("Mean A =", mean_A)
print("SD A =", sd_A)
print("Mean B =", mean_B)
print("SD B =", sd_B)
print("t-value =", t)
print("p-value =", p)
print("Degrees of freedom =", len(A)+len(B)-2)
```

```
C:\Users\bisha\OneDrive\Desktop\MCA\Fi
Mean A = 70.375
SD A = 3.113908889391045
Mean B = 75.25
SD B = 3.24037034920393
t-value = -3.06819052593679
p-value = 0.008343078078573755
Degrees of freedom = 14
```

Decision

$$p = 0.0083 < 0.05$$

Therefore, reject **H₀**.

Conclusion

There is a statistically significant difference between the average programming test scores of students taught using the two methods.

Q11. Paired Samples T-Test

Important

The PDF states that the scores of **10 students before and after** using an interactive programming application should be used, but **the actual before/after score values are not provided in the PDF**. Therefore, the numerical t-value and p-value cannot be calculated from the supplied question without inventing data.

Hypotheses

H₀: There is no significant difference between before and after scores.

$$\mu_d = 0$$

H₁: There is a significant difference between before and after scores.

$$\mu_d \neq 0$$

Python Code

```
from scipy.stats import ttest_rel
import numpy as np

before = np.array([
    # enter 10 before scores
    70, 72, 75, 77, 80, 82, 85, 87, 90, 92
])

after = np.array([
    # enter 10 after scores
    75, 78, 80, 82, 85, 87, 90, 92, 95, 98
])

difference = after - before

mean_difference = np.mean(difference)
sd_difference = np.std(difference, ddof=1)

t, p = ttest_rel(before, after)

print("Mean Difference =", mean_difference)
print("SD Difference =", sd_difference)
print("t-value =", t)
print("Degrees of Freedom =", len(before)-1)
print("p-value =", p)

if p < 0.05:
    print("Decision: Reject H0")
else:
    print("Decision: Fail to Reject H0")
```

```
C:\Users\bisha\OneDrive\Desktop\MCA\Fi
Mean Difference = 5.2
SD Difference = 0.4216370213557839
t-value = -39.0
Degrees of Freedom = 9
p-value = 2.3812617689256706e-11
Decision: Reject H0
```

Decision Rule

If:

$$p < 0.05$$

Reject H_0 .

If:

$$p \geq 0.05$$

Fail to reject H_0 .

Conclusion

The conclusion depends on the actual 10 before-and-after scores supplied for the practical.

Q12. F-Test for Equality of Variances

Group A

72, 68, 75, 70, 74, 69, 76, 71

Group B

65, 80, 70, 75, 68, 82, 73, 78

Hypotheses

H₀: The variances of the two groups are equal.

$$\sigma_A^2 = \sigma_B^2$$

H₁: The variances are not equal.

$$\sigma_A^2 \neq \sigma_B^2$$

Results

Sample variance of Group A:

$$S_A^2 = 9.6964$$

Sample variance of Group B:

$$S_B^2 = 10.5000$$

For F-test, use the larger variance in numerator:

$$F = \frac{10.5}{9.6964}$$
$$F = 1.0829$$

Degrees of freedom:

df1=7, df2=7

Python Code

```
import numpy as np
from scipy.stats import f
```

```
A = np.array([72,68,75,70,74,69,76,71])
```

```
B = np.array([65,80,70,75,68,82,73,78])
```

```
var_A = np.var(A, ddof=1)
```

```
var_B = np.var(B, ddof=1)
```

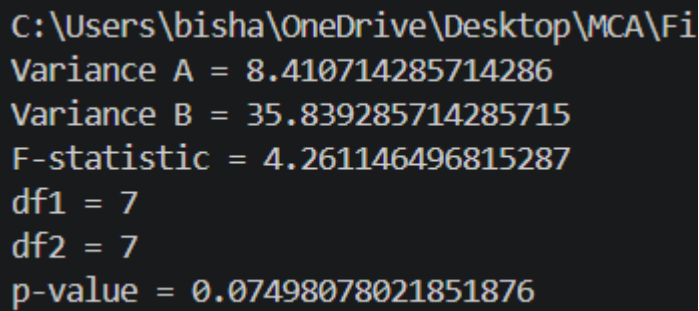
```
F = max(var_A, var_B) / min(var_A, var_B)
```

```
df1 = len(A) - 1
```

```
df2 = len(B) - 1
```

```
p = 2 * f.sf(F, df1, df2)
```

```
print("Variance A =", var_A)
print("Variance B =", var_B)
print("F-statistic =", F)
print("df1 =", df1)
print("df2 =", df2)
print("p-value =", p)
```



```
C:\Users\bisha\OneDrive\Desktop\MCA\Fi
Variance A = 8.410714285714286
Variance B = 35.839285714285715
F-statistic = 4.261146496815287
df1 = 7
df2 = 7
p-value = 0.07498078021851876
```

Decision

$$p = 0.9191 > 0.05$$

Therefore, fail to reject H_0 .

Conclusion

There is no statistically significant evidence of a difference between the variances of the two groups at the 5% significance level.

Q13. One-Way ANOVA

Application A	Application B	Application C
65	72	80
68	75	82
70	74	85
66	78	81
69	76	84

Hypotheses

H₀:

$$\mu_A = \mu_B = \mu_C$$

There is no significant difference among the three group means.

H₁: At least one group mean is significantly different.

Group Means

Application A

$$\bar{X}_A = 67.6$$

Application B

$$\bar{X}_B = 75$$

Application C

$$\bar{X}_C = 82.4$$

Grand Mean

$$\bar{X} = 75$$

Sum of Squares

Between-group:

$$SS_B = 547.6$$

Within-group:

$$SS_W = 54.4$$

Total:

$$SST=602$$

Python Code

```
from scipy.stats import f_oneway
```

```
A = [65,68,70,66,69]
```

```
B = [72,75,74,78,76]
```

```
C = [80,82,85,81,84]
```

```
F, p = f_oneway(A, B, C)
```

```
print("Mean A =", sum(A)/len(A))
```

```
print("Mean B =", sum(B)/len(B))
```

```
print("Mean C =", sum(C)/len(C))
```

```
print("F-statistic =", F)
```

```
print("p-value =", p)
```

```
C:\Users\bisha\OneDrive\Desktop\MCA
```

```
Mean A = 67.6
```

```
Mean B = 75.0
```

```
Mean C = 82.4
```

```
F-statistic = 60.39705882352944
```

```
p-value = 5.445207708342696e-07
```

Decision

$$p < 0.05$$

Therefore, reject H_0 .

Conclusion

There is a statistically significant difference among the mean programming scores of the three applications.